

Yoshita Lakshmi Mitta

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PROFESSIONAL SUMMARY

Computer Science undergraduate with strong foundations in data structures, algorithms, and full-stack software development, demonstrated through solving 300+ DSA problems. Passionate about building scalable, customer-centric software systems with hands-on experience developing AI-powered backend applications and distributed solutions.

EDUCATION

Rajiv Gandhi University of Knowledge Technologies (IIIT RK Valley)
Bachelor of Technology – Computer Science (CGPA: 9.2)

Kadapa, AP, India
2023 – 2027

SKILLS

Programming Languages: Python, JavaScript, C++, SQL

Frontend: React.js, HTML, CSS

Backend: Node.js, Express.js, FastAPI, LangChain, REST APIs

Databases: PostgreSQL, MongoDB, Supabase, Vector Databases

AI & GenAI: Large Language Models (LLMs), Conversational AI, Retrieval-Augmented Generation (RAG), Prompt Engineering, Semantic Search, Embeddings

Cloud & Deployment: AWS, Vercel, Firebase

Core CS: DBMS, OOPS, OS, CN, DSA

Tools: Git, GitHub, Postman, Chrome DevTools

PROJECTS

AuraMom – AI-Powered Maternal Wellness Platform | [GitHub](#) | [Live-Demo](#)

- **Tech Stack:** React.js, FastAPI, Gemini API, Hume AI TTS, Supabase
- Built a **personalized maternal wellness platform** delivering **pregnancy-stage-specific nutrition guidance**, **AI-powered meditation sessions**, and **daily wellness recommendations**.
- Engineered an **end-to-end AI pipeline** leveraging **Gemini API** and **Hume AI TTS** to analyze **user mood**, dynamically generating **context-aware meditation scripts** and **natural voice-guided audio sessions**.

AI Brain — Document Intelligence System | [GitHub](#)

- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, LangChain, Gemini 2.5 Flash, Qdrant, Nomic Embeddings, Unstructured API
- Built a **scalable RAG-powered Document Intelligence System** enabling **conversational PDF querying** through **vector embeddings**, **intelligent chunking**, **hybrid semantic-keyword retrieval**, improving search relevance by **~40%**.
- Engineered **production-grade AI pipelines** with **batched embedding generation**, **session-aware conversational memory**, **multi-user chat history**, and optimized backend workflows for **low-latency**, **context-aware document interactions**.

CareRoute AI — Real-Time AI Emergency Dispatch Engine | [GitHub](#)

- **Tech Stack:** React.js, Node.js, Express.js, Supabase, LiveKit, Leaflet.js, Google Gemini 1.5 Flash
- Built a **real-time AI emergency dispatch platform** that reduced hospital discovery and emergency coordination time from **45 minutes to under 30 seconds** by integrating **Google Gemini**, **LiveKit WebRTC**, and **AI-driven emergency triage**.
- Designed a **multi-constraint patient-to-hospital matching engine** enabling intelligent hospital selection and real-time emergency resource allocation.
- Engineered a **real-time dispatch infrastructure** with **Supabase WebSockets**, **lock-token based resource reservation**, **live ambulance GPS tracking** using **Leaflet.js**, and **RBAC-secured multi-user workflows** for Patients, Hospital Admins, and Ambulance Drivers.

EXPERIENCE

Web Development Mentor

Sep. 2024 – Nov. 2024

Student Recreation Center (SRC)

- * Mentored 30+ students in frontend development and explained technical concepts through interactive coding sessions, strengthening communication and problem-solving skills.